

Negative real zeros of a rising-factorial transform

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Abstract

We prove that a palindromic polynomial with positive constant term and only negative real zeros is carried, by a particular difference of rising-factorial transforms, either to zero or to a polynomial of degree two less with only negative real zeros. The proof factors the reciprocal root pairs, differentiates the resulting gamma polynomial, and applies an elementary interlacing lemma for a factorial basis. This yields the coefficient-pairing assertion for every polynomial of degree greater than two with only negative real roots. An additional Hadamard product with a binomial polynomial yields the positive-endpoint formulation of Karp's 2012 Conjecture 3. If zero endpoints are allowed in a finite Polya frequency sequence, the corresponding limit statement has nonpositive zeros, and we give an explicit example showing why strict negativity requires qualification.

1. Statements and scope

Throughout, $(x)_0 = 1$ and

$$(x)_k = x(x+1)\cdots(x+k-1) \quad (k \geq 1)$$

are rising factorials. A nonzero constant polynomial is regarded as having no zeros. The zero polynomial is treated separately. All polynomials below have real coefficients, and zeros are counted with multiplicity.

Theorem 1 (coefficient pairing). Let $n \geq 3$ and let

$$p(z) = \sum_{k=0}^n a_k z^k$$

have degree n and only strictly negative real zeros. Then

$$P_n(x) = \sum_{k=0}^n a_k a_{n-k} [(x)_k (x)_{n-k} - (x+1)_k (x-1)_{n-k}] \quad (1)$$

has degree $n-2$ and only strictly negative real zeros.

Karp [K12, Conjecture 3 and equation (7)] considers

$$\begin{aligned}\mathcal{K}_n[f](x) &:= Q_n^{1,1}(x-1) \\ &= \sum_{k=0}^n \binom{n}{k} f_k f_{n-k} [(x)_k (x)_{n-k} - (x+1)_k (x-1)_{n-k}].\end{aligned}\tag{2}$$

A finite sequence is PF_∞ when its associated Toeplitz matrix has all minors nonnegative. By the Aissen--Schoenberg--Whitney theorem [ASW52], this is equivalent to its nonzero generating polynomial having nonnegative coefficients and only real nonpositive zeros.

Theorem 2 (positive-endpoint form of Karp's Conjecture 3). Let $n \geq 3$ and let $(f_0, \dots, f_n)$ be PF_∞ with $f_0 f_n > 0$. Then $\mathcal{K}_n[f]$ has degree $n-2$ and only strictly negative real zeros.

The condition $f_0 f_n > 0$ makes the scope explicit: the input has degree exactly n and no zero at the origin. Section 6 treats sequences with zero endpoints. No assertion of historical priority is needed for either theorem; the implication for Karp's formulation is proved in Section 5.

2. A factorial-transform lemma

For $d \geq 0$, $\alpha > 0$, and $h(t) = \sum_{r=0}^d h_r t^r$, define

$$T_{d,\alpha} h(x) = \sum_{r=0}^d h_r (x)_r (x+r+\alpha)_{d-r}.\tag{3}$$

Lemma 3. Suppose h is nonzero, has nonnegative coefficients, $h_0 > 0$, and has only strictly negative real zeros, with $\deg h \leq d$. Then $T_{d,\alpha} h$ has degree d and only strictly negative real zeros. Its consecutive zeros are separated by at least one.

Proof. Normalize $h_0 = 1$; this changes neither set of zeros. Factor

$$h(t) = \prod_{j=1}^d (1 + \mu_j t), \quad \mu_j \geq 0,$$

padding with zero values of μ_j if $\deg h < d$. Put $h_i(t) = \prod_{j=1}^i (1 + \mu_j t)$ and $Q_i = T_{i,\alpha} h_i$. Then $Q_0 = 1$ and

$$Q_i(x) = (x + \alpha + i - 1) Q_{i-1}(x) + \mu_i x Q_{i-1}(x+1).\tag{4}$$

Indeed, for the basis elements $B_{d,r}(x) = (x)_r (x+r+\alpha)_{d-r}$,

$$B_{d+1,r}(x) = (x + \alpha + d) B_{d,r}(x), \quad B_{d+1,r+1}(x) = x B_{d,r}(x+1),$$

which gives (4) by multiplying h_{i-1} by $1 + \mu_i t$.

Assume first that all $\mu_j > 0$. We prove by induction that Q_i has i simple zeros in $(-\alpha - i + 1, 0)$, with consecutive zeros separated by more than one. For $i = 1$,

$$Q_1(x) = (1 + \mu_1)x + \alpha,$$

so the assertion holds. Suppose it holds for $i-1$, with zeros $r_1 < \dots < r_{i-1}$. The two polynomials

$$A(x) = (x + \alpha + i - 1)Q_{i-1}(x), \quad B(x) = xQ_{i-1}(x + 1)$$

have positive leading coefficients, and their zeros alternate as follows:

$$-\alpha - i + 1 < r_1 - 1 < r_1 < r_2 - 1 < \dots < r_{i-1} - 1 < r_{i-1} < 0. \quad (5)$$

The first inequality follows from $r_1 > -\alpha - i + 2$; the others follow from the induction hypothesis. Write $\xi_1 = -\alpha - i + 1$, $\xi_j = r_{j-1}$ for $2 \leq j \leq i$, $\eta_j = r_j - 1$ for $1 \leq j < i$, and $\eta_i = 0$. Thus $\xi_1 < \eta_1 < \dots < \xi_i < \eta_i$, where the ξ_j are the roots of A and the η_j the roots of B .

At ξ_j , the sign of $A + \mu_i B$ is $(-1)^{i-j+1}$, because $i - j + 1$ roots of B lie to its right. At η_j , its sign is $(-1)^{i-j}$, because $i - j$ roots of A lie to its right. Consequently $Q_i = A + \mu_i B$ has a root in each of the i intervals (ξ_j, η_j) . Its degree is i , so these are all its roots. If they are $s_1 < \dots < s_i$, then

$$s_j < r_j - 1 < r_j < s_{j+1} \quad (1 \leq j < i).$$

Hence $s_{j+1} - s_j > 1$, and the induction is complete.

For general $\mu_j \geq 0$, replace them by $\mu_j + \delta$ and let $\delta \downarrow 0$. The leading coefficient of (3) is $\sum_r h_r = h(1) > 0$, so the degree remains d in the limit. Continuity of roots gives real nonpositive zeros and spacing at least one. Finally,

$$T_{d,\alpha}h(0) = h_0(\alpha)_d > 0,$$

so none of those zeros is zero. This proves the lemma. $\square$

3. Palindromic polynomials and the difference transform

For a coefficient vector $c = (c_0, \dots, c_n)$ define

$$\mathcal{D}_n[c](x) = \sum_{k=0}^n c_k [(x)_k(x)_{n-k} - (x+1)_k(x-1)_{n-k}]. \quad (6)$$

Theorem 4. Let $n \geq 2$ and let $C(z) = \sum_{k=0}^n c_k z^k$ have degree n , positive constant term, only strictly negative real zeros, and $c_k = c_{n-k}$. If $C(z) = c_0(1+z)^n$, then $\mathcal{D}_n[c] \equiv 0$. Otherwise $\mathcal{D}_n[c]$ has degree $n-2$ and only strictly negative real zeros.

Proof. Write $n = 2m + \varepsilon$, where $\varepsilon \in \{0, 1\}$. Reciprocal symmetry pairs every root with its reciprocal. A pair $-r, -r^{-1}$, $r > 0$, contributes

$$(z+r)(z+r^{-1}) = (1+z)^2 + \lambda z, \quad \lambda = r + r^{-1} - 2 \geq 0.$$

Any roots at -1 can be paired except for one when n is odd. It follows that

$$\begin{aligned} C(z) &= c_0(1+z)^\varepsilon \prod_{i=1}^m ((1+z)^2 + \lambda_i z) \\ &= c_0 \sum_{j=0}^m \gamma_j z^j (1+z)^{n-2j}, \end{aligned} \quad (7)$$

where

$$\Gamma(t) := \sum_{j=0}^m \gamma_j t^j = \prod_{i=1}^m (1 + \lambda_i t). \quad (8)$$

Define $F_C(x, y) = \sum_{k=0}^n c_k(x)_k(y)_{n-k}$. For each summand in (7), the rising-factorial Vandermonde identity gives

$$\begin{aligned} & \sum_{\ell=0}^{n-2j} \binom{n-2j}{\ell} (x)_{j+\ell}(y)_{n-j-\ell} \\ &= (x)_j(y)_j \sum_{\ell=0}^{n-2j} \binom{n-2j}{\ell} (x+j)_\ell(y+j)_{n-2j-\ell} \\ &= (x)_j(y)_j(x+y+2j)_{n-2j}. \end{aligned}$$

Therefore

$$F_C(x, y) = c_0 \sum_{j=0}^m \gamma_j (x)_j (y)_j (x+y+2j)_{n-2j}. \quad (9)$$

For $j \geq 1$ we have the polynomial identity

$$\begin{aligned} & (x)_j^2 - (x+1)_j(x-1)_j \\ &= (x)_{j-1}(x+1)_{j-1}[x(x+j-1) - (x-1)(x+j)] \\ &= j(x)_{j-1}(x+1)_{j-1}. \end{aligned} \quad (10)$$

Taking $F_C(x, x) - F_C(x+1, x-1)$ in (9) cancels the $j=0$ term and yields

$$\mathcal{D}_n[c](x) = c_0 \sum_{j=1}^m j \gamma_j (x)_{j-1}(x+1)_{j-1}(2x+2j)_{n-2j}. \quad (11)$$

Now split the last factor using

$$(2u)_{2v} = 4^v (u)_v (u + \frac{1}{2})_v, \quad (2u)_{2v+1} = 2 \cdot 4^v (u)_{v+1} (u + \frac{1}{2})_v.$$

Since

$$(x+1)_{j-1}(x+j)_{m-j+\varepsilon} = (x+1)_{m-1+\varepsilon},$$

we obtain

$$\mathcal{D}_n[c](x) = 2^\varepsilon c_0 (x+1)_{m-1+\varepsilon} R(x), \quad (12)$$

where

$$R(x) = \sum_{r=0}^{m-1} (r+1) \gamma_{r+1} 4^{m-1-r} (x)_r (x+r+\frac{3}{2})_{m-1-r}. \quad (13)$$

If all $\lambda_i = 0$, then $\Gamma = 1$ and (11) vanishes. Otherwise Γ is nonconstant, $\Gamma'(0) = \sum_i \lambda_i > 0$, and Rolle's theorem implies that Γ' has only strictly negative real zeros (or is a positive constant). Set

$$h(t) = 4^{m-1} \Gamma'(t/4). \quad (14)$$

This polynomial has nonnegative coefficients, positive constant term, and degree at most $m - 1$. Equation (13) is exactly

$$R = T_{m-1, 3/2} h.$$

Lemma 3 shows that R has degree $m - 1$ and only strictly negative real zeros. The factor $(x + 1)_{m-1+\varepsilon}$ has the same zero property. Their degrees add to $2m - 2 + \varepsilon = n - 2$, proving the theorem. $\square$

In particular, (12) displays the fixed negative integer roots. The remaining factor has simple negative zeros separated by at least one. This does not assert that all roots of the full product are distinct: a root of R may coincide with one of the fixed integer roots.

4. Proof of the coefficient-pairing theorem

We use the classical Hadamard-product theorem: if two real polynomials have nonnegative coefficients and only real nonpositive zeros, then their nonzero coefficientwise product also has only real nonpositive zeros. This is the consequence of Malo's theorem recalled in [GW96, p. 798]; see also [GW96, Theorem 4(a)]. A positive constant term excludes a zero at the origin.

Replace p by $-p$ if necessary so that its coefficients are positive. Its reciprocal

$$p^*(z) = z^n p(1/z) = \sum_{k=0}^n a_{n-k} z^k$$

also has only negative real zeros. Thus

$$C(z) = p \odot p^*(z) = \sum_{k=0}^n a_k a_{n-k} z^k$$

has only negative real zeros and is palindromic, where $\odot$ means coefficientwise product.

It remains to exclude $C = c_0(1 + z)^n$. Write $p(z) = a_n \prod_{i=1}^n (z + s_i)$ with $s_i > 0$. Cauchy--Schwarz gives

$$\frac{c_1}{c_0} = \frac{a_1 a_{n-1}}{a_0 a_n} = \left(\sum_{i=1}^n s_i \right) \left(\sum_{i=1}^n s_i^{-1} \right) \geq n^2 > n. \quad (15)$$

But the corresponding ratio for $c_0(1 + z)^n$ is n . The nonzero case of Theorem 4 applies, proving Theorem 1. $\square$

For completeness, the formula gives $P_0 = P_1 = 0$ and $P_2 = a_1^2 - 2a_0 a_2 > 0$. The assertion for $n \geq 3$ therefore concerns a nonconstant polynomial throughout.

5. The binomial factor in Karp's conjecture

Let $F(z) = \sum_{k=0}^n f_k z^k$ satisfy the hypotheses of Theorem 2. By [ASW52], F has only strictly negative real zeros. Define

$$C_K(z) = \sum_{k=0}^n \binom{n}{k} f_k f_{n-k} z^k.$$

The extra binomial factor is handled by a third polynomial in the Hadamard product:

$$C_K = F \odot (z^n F(1/z)) \odot (1+z)^n. \quad (16)$$

Applying the Hadamard-product theorem twice proves that C_K has only strictly negative real zeros. It is palindromic, has degree n , and has constant term $f_0 f_n > 0$. Applying (15) to F gives

$$\frac{[z]C_K(z)}{C_K(0)} = n \frac{f_1 f_{n-1}}{f_0 f_n} \geq n^3 > n.$$

Hence $C_K \neq C_K(0)(1+z)^n$. Theorem 4, applied to the coefficients of C_K , proves that (2) has degree $n-2$ and only strictly negative real zeros. This proves Theorem 2. $\square$

Equation (16) is essential to the comparison: the original coefficient-pairing formula (1) and Karp's formula (2) differ by $\binom{n}{k}$. No identification of their coefficient sequences without this extra step is being made. The result establishes the conjectured conclusion when both endpoints are positive; the distinction between negative and nonpositive roots at zero endpoints is addressed next.

6. Zero endpoints

Proposition 5. Let $n \geq 3$. If $(f_0, \dots, f_n)$ is a nonzero finite PF_∞ sequence, allowing zero endpoints, then $\mathcal{K}_n[f]$ is either zero or has only real nonpositive zeros.

Proof. Factor its generating polynomial as

$$F(z) = Az^s \prod_{i=1}^{D-s} (z + u_i), \quad A > 0, \quad u_i > 0, \quad D \leq n.$$

For $\delta > 0$ set

$$F_\delta(z) = A(z + \delta)^s \prod_{i=1}^{D-s} (z + u_i)(1 + \delta z)^{n-D}.$$

Each F_δ has degree n , positive endpoints, and only strictly negative real zeros. Its coefficients converge to those of F , padded through degree n . Theorem 2 applies to $\mathcal{K}_n[F_\delta]$, whose coefficients converge to those of $\mathcal{K}_n[f]$. A nonzero coefficientwise limit of polynomials of bounded degree with all zeros in $(-\infty, 0]$ again has all zeros in that closed set. This follows, for example, by continuity of zeros near any finite zero of the nonzero limit. The proposition follows. $\square$

Strict negativity need not survive. Take $n = 4$ and

$$(f_0, f_1, f_2, f_3, f_4) = (1, 2, 1, 0, 0), \quad F(z) = (1+z)^2.$$

This is a PF_∞ sequence, but only $k = 2$ contributes to (2), so

$$\mathcal{K}_4[f](x) = 6[(x)_2^2 - (x+1)_2(x-1)_2] = 12x(x+1).$$

If degree exactly n is required but a zero constant coefficient is allowed, $(0, 0, 1, 2, 1)$ gives the same example. Thus the positive-endpoint hypothesis in Theorem 2 is substantive for the stated strict conclusion. Karp's use of finite PF_∞ sequences should be read with its endpoint convention made explicit; no counterexample to the positive-endpoint form is asserted here.

7. Verification and provenance

The proof uses the classical Hadamard-product theorem and, for the PF_∞ formulation, the Aissen-Schoenberg-Whitney characterization. The factorial-transform lemma, reciprocal-pair reduction, and algebraic identities are proved in this manuscript. The supplementary SymPy script checks the identities, transform recurrence, sample outputs with exact real-root counts, and the zero-endpoint example. Those finite checks are supporting evidence and do not replace the proof for arbitrary degree.

This manuscript was prepared with OpenAI Codex from the author's problem statement and publication instructions. The drafting assistant also performed the algebraic audit. An off-by-one endpoint in an earlier conversational version of the interlacing argument has been corrected: the leftmost root in (5) is exactly $-\alpha - i + 1$. No independent human review, journal acceptance, or exhaustive literature determination is claimed.

References

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